

Best Practices for Scientific Writing in Introductions

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Purpose and Goals of Scientific Introductions

Scientific introductions serve three main purposes: establishing the importance and need for research, demonstrating how the study fills knowledge gaps, and previewing what readers will learn. They should move from broad context to specific research problems while clearly articulating the study's novelty and significance. (5 sources)

The primary purpose of a scientific introduction is multifaceted, serving to establish the foundation for the entire research paper. The introduction has three main goals: to establish the need and importance of your research, to indicate how you have filled the knowledge gap in your field, and to give your readers a hint of what they will learn when reading your paper ([Delvin et al., 2014](#)). This framework requires authors to outline the problem to be investigated, why the study is novel, why it is important, and how it builds on a body of existing work and knowledge ([Troy et al., 2021](#)).

The introduction follows a strategic organizational pattern that moves from general to specific. It begins broadly by introducing the area of interest, narrows to identify the specific focus and gap in knowledge, and finally ends with the aim of the present study, seamlessly leading to the methods and results sections ([Sefcik et al., 2024](#)). This structure ensures that the introduction provides the salient background for readers to understand why the research question is important, highlighting both the need for the present study and how the anticipated findings might impact practice ([Evans et al., 2024](#)).

A well-constructed introduction defines the problem to be addressed, identifies what is known and unknown about the problem, and states the study purpose ([Sefcik et al., 2024](#)). The content should include relevant background literature and theories to be studied, setting up the techniques, results, and discussion that will connect the findings to the topic's broader information ([Akintobi et al., 2023](#)).

Evidence

(Delvin et al., 2014)

[How to Write a Scientific Paper: Practical Guidelines](#)

"The introduction has 3 main goals: to establish the need and importance of your research, to indicate how you have filled the knowledge gap in your field and to give your readers a hint of what they will learn when reading your paper. To fulfil these goals, a four-part introduction consisting of a background statement, a problem statement, an activity statement and a forecasting statement, is best suited. Poorly defined background information and problem setting are the 2 most common weaknesses encountered in introductions. They stem from the false perception that peer readers know what the issue is and why the study to solve it is necessary. Although not a strict rule, the introduction in clinical science journals should target only references needed to establish the rationale for the study and the research protocol."

(Troy et al., 2021)

[Creating a Course on Scientific Writing and Oral Presentation in English for Engineering and Science Graduate Students at an Elite Chinese University](#)

"For the Introduction section, the author is expected to outline the problem to be investigated, why the study is novel, why it is important, and how it builds on a body of existing work and knowledge."

(Sefcik et al., 2024)

[Writing Manuscript "Bookends": Strategies to Effectively Frame Science.](#)

"The introduction defines the problem to be addressed, identifies what is known and unknown about the problem, and states the study purpose. It begins broadly by introducing the area of interest, narrows to identify the specific focus and gap in knowledge, and finally ends with the aim of the present study, seamlessly leading to the methods and results sections."

(Evans et al., 2024)

[Clinical Research Primer for Medical Students: Overview and Illustrative Experiences](#)

"The Introduction provides the salient background for readers to understand why the research question is important, highlighting both the need for the present study and how the anticipated findings might impact practice, with a target length of one to three paragraphs that are tightly focused on the study question."

(Akintobi et al., 2023)

[Writing a Good Research Paper](#)

"The introduction to a specific issue with relevant background literature and theories to be studied, a list and explanation of the techniques, results, and a discussion that connects the results to the topic's broader information should all be included in a scientific paper's main text."

Structural Frameworks for Introduction Organization

Scientific introductions should follow established structural frameworks like the three-move approach or funnel/inverted pyramid structure, moving from broad context to specific research questions. These frameworks typically organize content into 3-4 paragraphs covering general background, literature review, knowledge gaps, and study objectives. (15 sources)

Several well-established structural frameworks guide the organization of effective scientific introductions. The most widely recommended approach is the **three-move structure**, which provides clear traffic signs that lead readers through your ideas ([Kallestinova, 2011](#)). This framework consists of: (1) establishing the territory by placing research in context and highlighting topic importance, (2) finding the niche by outlining the research problem scope, and (3) occupying the niche by explaining the research and highlighting the paper's significance ([Kallestinova, 2011](#)).

Another common organizational approach is the **inverted pyramid or funnel structure**, which starts wide with broad context and narrows to specific research questions ([Vengadasalam, 2020](#)) ([Turbek et al., 2016](#)). This funnel approach should provide enough background that someone outside the field can understand the topic, then narrow the scope until reaching the specific question being addressed ([Turbek et al., 2016](#)).

The **four-paragraph framework** offers a more detailed structure, with each paragraph addressing specific questions in a funnel-shaped manner from generalized to specific details ([Aga et al., 2022](#)) ([Cals et al., 2013](#)) ([Ahlstrom, 2017](#)) ([Mensh et al., 2016](#)) ([Behzadi et al., 2021](#)) ([Shokeir, 2013](#)). This structure follows the **"three Cs" scheme**: context (what has been published), content (what you will do about the problem), and conclusion (how it will be done) ([Aga et al., 2022](#)) ([Mensh et al., 2016](#)) ([Fried et al., 2019](#)).

A more recent framework is the **"problem, gap, hook" method**, which structures the narrative around three key elements ([Wilson et al., 2024](#)) ([Gottlieb et al., 2018](#)) ([Lingard, 2015](#)). This approach identifies a significant problem and its context, establishes a gap in current knowledge, and hooks readers by convincing them that filling this gap has important consequences ([Wilson et al., 2024](#)).

For practical implementation, introductions should comprise about 10-15% of the paper's word count and ideally stay below 500 words ([Aga et al., 2022](#)) ([Cals et al., 2013](#)) ([Carandini, 2022](#)). The specific components

should include: (1) scenario with general field information, (2) background of relevant previous studies, (3) knowledge gap identification, and (4) concise research question or objective ([Silveira et al., 2022](#)).

Evidence

(Kallestinova, 2011)

[How to Write Your First Research Paper](#)

"The best way to structure your introduction is to follow the three-move approach shown in Table 3. The moves and information from your outline can help to create your Introduction efficiently and without missing steps. These moves are traffic signs that lead the reader through the road of your ideas. Each move plays an important role in your paper and should be presented with deep thought and care. When you establish the territory, you place your research in context and highlight the importance of your research topic. By finding the niche, you outline the scope of your research problem and enter the scientific dialogue. The final move, "occupying the niche," is where you explain your research in a nutshell and highlight your paper's significance."

(Vengadasalam, 2020)

[Publish or Perish! Sharing Best practices for a Writing Instructor Led "Writing for Publication" Course](#)

"I coach them to write introductions as inverted pyramids"

(Turbek et al., 2016)

[Scientific Writing Made Easy: A Step-by-Step Guide to Undergraduate Writing in the Biological Sciences](#)

"Think of your Introduction as the beginning of a funnel: Start wide to put your research into a broad context that someone outside of the field would understand, and then narrow the scope until you reach the specific question that you are trying to answer (Fig. 1; Schimel 2012). Clearly state the wider implications of your work for the field of study, or, if relevant, any societal impacts it may have, and provide enough background information that the reader can understand your topic. Perform a thorough sweep of the literature; however, do not parrot everything you find. Background information should only include material that is directly relevant to your research and fits into your story; it does not need to contain an entire history of the field of interest."

(Aga et al., 2022)

[Essential Guide to Manuscript Writing for Academic Dummies: An Editor's Perspective](#)

"An effective introduction usually constitutes about 10-15% of the paper's word count (Cals et al., 2013)"

"It should generally be made up of four paragraphs, each dedicated to cover four specific questions in a funnel-shaped manner, traversing from a more generalized view to the specific details (Figure 6) (Mensh et al., 2016)(Behzadi et al., 2021)(Shokeir, 2013) (Ohwovoriole, 2011)(Ahlstrom, 2017)"

"Introduction is regarded as the start of the storyline of manuscript, and hence, the three Cs' scheme (Figure 5) becomes more relevant while writing it: the context in terms of what has been published on the current idea/problem around the world, content as to what you are going to do about the problem in hand (rationale), and conclusion as to how it is going to be done (specific objective of the study) (Mensh et al., 2016)(Fried et al., 2019)"

"Introduction should never be written like a review, any details, contexts, and comparisons should be dealt within the discussion part [16]."

(Cals et al., 2013)

[Effective writing and publishing scientific papers, part III: introduction.](#)

"Check if the introduction has a funnel shape with clear sections on B general background (what is this all about?); B what is known and what is unknown about this specific subject (why was this study needed, and why is it important?); B primary research question (what did we want to know?); and B study aim and design (what did we do to answer the research question?). Look at the length of the introduction (maximum 10-15% of the total word count). Determine if the introduction is the start of the story line of your paper by looking at your outline (skeleton). Ask yourself, "Will this introduction sell my paper to editors, reviewers, readers, and the media?"
1. What you should know"

(Ahlstrom, 2017)

[How to Publish in Academic Journals: Writing a Strong and Organized Introduction Section](#)

"Publishing in quality academic journals is challenging. Authors who want to improve their chances of publishing in management and allied business and social science journals can save themselves much time and frustration by ensuring that manuscripts are consistent with the journal's aims and scope and what the field requires in terms of addressing unanswered research questions or

improvements to current theory and evidence. It is well-understood if a manuscript lacks theoretical grounding or makes significant methodological or research design mistakes, it will likely be rejected. Researchers in the social sciences are typically well-trained in methods, statistical analysis, and research design. But many scholars have much less training on the situating, motivating, and organizing of manuscripts, particularly in the all-important introduction of the paper. Oftentimes, an author may face rejection of his or her manuscript not because of bad data or methods, but because of major framing and organizational issues with the paper, as well as a lack of clear contributions. These problems are addressed within the context of writing a clear research question and introduction section, which form the basis for the overall organization of the paper. Numerous helpful sources are also provided."

(Mensh et al., 2016)

[Ten simple rules for structuring papers](#)

"Good scientific writing is essential to career development and to the progress of science. A well-structured manuscript allows readers and reviewers to get excited about the subject matter, to understand and verify the paper's contributions, and to integrate these contributions into a broader context. However, many scientists struggle with producing high-quality manuscripts and typically get little training in paper writing. Focusing on how readers consume information, we present a set of 10 simple rules to help you get across the main idea of your paper. These rules are designed to make your paper more influential and the process of writing more efficient and pleasurable."

(Behzadi et al., 2021)

[Writing a strong scientific paper in medicine and the biomedical sciences: a checklist and recommendations for early career researchers](#)

"Scientific writing is an important skill in both academia and clinical practice. The skills for writing a strong scientific paper are necessary for researchers (comprising academic staff and health-care professionals). The process of a scientific research will be completed by reporting the obtained results in the form of a strong scholarly publication. Therefore, an insufficiency in scientific writing skills may lead to consequential rejections. This feature results in undesirable impact for their academic careers, promotions and credits. Although there are different types of papers, the original article is normally the outcome of experimental/epidemiological research. On the one hand, scientific writing is part of the curricula for many medical programs. On the other hand, not every physician may have adequate knowledge on formulating research results for publication adequately. Hence, the present review aimed to introduce the details of creating a strong original article for publication (especially for novice or early career researchers)."

(Shokeir, 2013)

[How to write a medical original article: Advice from an Editor](#)

"Abstract Objectives: To provide guidelines for potential authors on how to increase the chances of their manuscript being accepted, with a review focusing on writing an original medical article. Methods: This review reflects the personal experience of the present author, who has extensive experience as an author, reviewer and editor. Results: To write an original article successfully, there are three essential requirements, the 'basic triad' of an original article. These are subjects worth reporting, knowledge of the basic structure of an article, and knowledge of the essential mechanics of good writing. This review details each of the three items. Conclusions: Writing, like every other art, cannot be learned wholly from books or lectures, but can be learned largely by experience. The best training is to start the task and persevere. The act of writing, like surgical techniques, must be learned the hard way, by practice and perseverance. Anyone can start writing but only a good writer can finish the task."

(Fried et al., 2019)

[How to Write an Effective Introduction](#)

"Ideally, the Introduction is an essential attention grabbing section of a research paper. If written correctly, the Introduction peaks the reader's interest as well as serves as a roadmap for the rest of the paper. An effective Introduction builds off related empirical research and demonstrates a gap in which the current study fills. Finally, the Introduction proposes the research question(s) which will be answered in subsequent sections of the paper. A strong Introduction also requires the use of a simple and well-organized format as well as the avoidance of common pitfalls."

(Wilson et al., 2024)

[Journal recommended guidelines for survey-based research](#)

"Follow the 'problem, gap, hook' method to write the introduction's narrative while citing the most relevant and current literature. Problem: What problem does the research aim to solve, and what is the context surrounding that problem? Gap: Establish a gap in the current knowledge or thinking about the problem and clearly demonstrate how the research meaningfully fills the identified gap. Hook: Hook readers by convincing them that filling the gap is of consequence. Emphasize the relevance and significance of the research, including its potential implications."

(Gottlieb et al., 2018)

[Scholarship in Emergency Medicine: A Primer for Junior Academics Part I: Writing and Publishing](#)

"The introduction should be formatted such that it presents a summary of the literature and how the study fits into the current understanding of the topic. This has been referred to as the problem/gap/hook heuristic. (Lingard, 2015) In this model, Lingard suggests that an introduction must do three things: identify a problem of significance to the reader; establish a gap in the current knowledge or understanding of the problem; and articulate a hook that convinces the reader of the importance of this. (Lingard, 2015)"

(Lingard, 2015)

[Joining a conversation: the problem/gap/hook heuristic](#)

"In the writer's craft section we offer simple tips to improve your writing in one of three areas: Energy, Clarity and Persuasiveness. Each entry focuses on a key writing feature or strategy, illustrates how it commonly goes wrong, teaches the grammatical underpinnings necessary to understand it and offers suggestions to wield it effectively. We encourage readers to share comments on or suggestions for this section on Twitter, using the hashtag: #how'syourwriting?"

(Carandini, 2022)

[Some Tips for Writing Science](#)

"Let's start here in the first paragraph, which should set up the broad area of inquiry and ideally open with a vivid sentence"

"The rest of the Introduction is there to pose specific questions in the context of the literature and as objects of debate. These questions are the pillars of the paper: pose them in the Introduction, answer them in Results, and discuss your answers in Discussion"

"When presenting the questions, beware of asserting that they are "interesting," "important," or "understudied." Let the reader conclude that they are interesting or important. As for "understudied," it makes a weak case: they may be understudied because unimportant"

"The last paragraph of the Introduction usually describes the main results. Mention the methods, especially if they are novel or provide new avenues into a problem, and summarize the take-home messages"

"Strive for brevity, ideally keeping the Introduction below 500 words."

(Silveira et al., 2022)

[Guide for scientific writing: how to avoid common mistakes in a scientific article](#)

"The introduction should be the last section to be written. Thus, authors will have greater objectivity in the relevant subjects to introduce the research topic. It should introduce the subject and provide reasons to justify the objective and relevance of the study. The introduction also presents the research problem and demonstrates knowledge gaps. A logical sequence should be established from the most general to the most specific aspects of the study (inverted triangle)"

"Mistakes in this section are more related to a very long introduction when the logical sequence of the text and the real problem to be highlighted by the research are lost. Another common mistake is to raise issues that will not be covered in the article. Excessively detailing what was observed in previous studies makes the text tiresome to read and loses its introductory characteristics. Citing current situations using old references is also a recurrent mistake in scientific studies"

"The introduction must contain: 1) Scenario (general information establishing the field of study) 2) Background (presentation of relevant previous studies) 3) Knowledge gap 4) Research question (concise definition of objective/hypothesis)"

Establishing Context and Research Territory

Establishing context requires placing your work within broader research while avoiding comprehensive literature reviews. Focus on directly relevant background that contextualizes your specific problem, demonstrates the current state of scientific understanding, and clearly shows how your research fits into the existing knowledge landscape. (6 sources)

The process of establishing context involves strategically placing your research within the broader scientific landscape while maintaining focus on your specific objectives. The introduction should place your work into the broader research context, then narrow your focus to identify specifically what you plan to do in the paper - your research objectives ([Nor, 2020](#)). This contextualization process requires presenting the necessary background and context of the research to provide readers with a clear synopsis of the study ([Abdullah, 2016](#)).

Effective context establishment involves **establishing the scope, context and significance of the research, summarizing current understanding and background information about the topic** ([Ezeifeka, 2021](#)). This

requires identifying an area of research and summarizing previous studies in a manner that lays a foundation for understanding the research problem ([Ezeifeka, 2021](#)). The background should reflect the state of scientific understanding and background of the research topic through the use of primary and secondary literature, relating it to the next logical research to be conducted ([Anderson et al., 2018](#)).

A critical principle in establishing context is **selectivity over comprehensiveness**. A common mistake authors make is presenting a full review of the literature in their field, when what is necessary is to mention only those aspects needed to contextualize the problem that your study is solving ([Nor, 2020](#)). Background information should only include material that is directly relevant to your research and fits into your story - it does not need to contain an entire history of the field of interest ([Turbek et al., 2016](#)).

The context should clearly state the wider implications of your work for the field of study, or if relevant, any societal impacts it may have, while providing enough background information that readers can understand your topic ([Turbek et al., 2016](#)). This contextualization should demonstrate relevance by showing that the study has certain significance at the present stage of scientific development, with clear arguments defending this position and attention to the degree of theoretical development of the problem ([Kondratieva et al., 2023](#)).

Evidence

(Nor, 2020)

[From Research and Practices into Journal Papers: Editor's Insight](#)

"Introduction is about placing your work into the broader research context, and then narrowing your focus to identify specifically what you plan to do in the paper -your research objectives. The broader context will be supported with the literature review. Therefore, introduction should include summaries of important and research which are relevant to your study. This is particularly important if there is an essential or groundbreaking study about the research problem or a key study that refutes or supports your research"

"However, a common mistake that authors make is to present a full review of the literature in their field. What is necessary is to mention those aspects that are needed to contextualize the problem that your study is solving."

(Abdullah, 2016)

[An Analysis of Cross-Discipline Research Article Introduction Structures through A Modified Create-A-Research-Space \(CARS\) Model](#)

"In writing academic text, presenting introduction is an attempt to provide the necessary background and context of the research (Noorzan & Page, 2012). The introduction of a research should be organized in a way to make sure that the readers can understand more about the synopsis of a research which would be explicitly explained later in the article. It depicts the research overview by giving a background of the study, putting forward the problems as significant factors of the study then describing how and why the problems will be solved"

"this section contextualizes a research study being presented in the relevant literature, claims its novelty and presents main features of the study (Swales, 1990)."

(Ezeifeka, 2021)

[Genre-Based Approach to the Analysis of Introductions in Doctoral Seminars of the Department of English, Unizik, Awka](#)

"establishing the scope, context and significance of the research, summarizing current understanding and background information about the topic"

"The writer must, first of all, ensure that an area of research is identified, that previous studies are summarized in a manner that lays a foundation for understanding the research problem; secondly, explain how the research problem addresses gaps in literature: insufficient treatment of the topic, missing links, imbalances and other deficiencies in literature; and finally, note the broader theoretical, empirical and/or policy contributions and implications of the research."

(Anderson et al., 2018)

[Development of a Uniform Approach to Writing and Grading of Laboratory Reports in Horticultural Science Courses](#)

"Through the use of primary and secondary literature, the writing reflects the state of the scientific understanding and background of the research topic, relating it to the next logical research to be conducted. The introduction leads the reader into the research objectives and hypothesis being tested."

(Turbek et al., 2016)

Scientific Writing Made Easy: A Step-by-Step Guide to Undergraduate Writing in the Biological Sciences

"Think of your Introduction as the beginning of a funnel: Start wide to put your research into a broad context that someone outside of the field would understand, and then narrow the scope until you reach the specific question that you are trying to answer (Fig. 1; Schimel 2012). Clearly state the wider implications of your work for the field of study, or, if relevant, any societal impacts it may have, and provide enough background information that the reader can understand your topic. Perform a thorough sweep of the literature; however, do not parrot everything you find. Background information should only include material that is directly relevant to your research and fits into your story; it does not need to contain an entire history of the field of interest."

(Kondratieva et al., 2023)

Final qualification work of a future engineer: Innovations and traditions

"The introduction of scientific work is the most important structural element of the study and represents its detailed plan. The introduction reflects the main characteristics of the work: relevance, historiography of the study, its object and subject, setting goals and objectives, determining the methods and material of the study, scientific novelty, theoretical and practical significance, approbation, provisions submitted for defense, the structure of the work. The total volume of the introduction text is 2-3 pages.

A study is topical if it has a certain significance at the present stage of the development of science. Justification of relevance is a mandatory requirement for any scientific work. To do this, it is necessary to clearly formulate the arguments in defense of one's position, as well as pay attention to the assessment of the degree of theoretical development of the problem and the need for its further scientific study.

Under the historiography of the work is meant the body of scientific sources on the basis of which it is built. The main result of the historiographic review is the identification of the state of scientific development of the topic."

Identifying and Articulating Research Problems and Gaps

Effective problem identification requires clearly defining what specific issue your research addresses and demonstrating meaningful gaps in current knowledge. The key is to move beyond simply stating what hasn't been studied to showing why filling particular knowledge gaps matters for scientific understanding or practical applications. (7 sources)

Identifying and articulating research problems involves a systematic approach that moves from recognizing broad issues to pinpointing specific knowledge deficiencies. The introduction must define the problem to be addressed, identify what is known and unknown about the problem, and state the study purpose ([Sefcik et al., 2024](#)). This process requires clearly explaining how the research problem addresses gaps in literature, including insufficient treatment of topics, missing links, imbalances and other deficiencies in literature ([Ezeifeke, 2021](#)).

The **problem-gap-hook framework** provides a systematic method for articulating research problems effectively. First, identify what problem the research aims to solve and the context surrounding that problem. Second, establish a gap in current knowledge or thinking about the problem and clearly demonstrate how the research meaningfully fills the identified gap. Finally, hook readers by convincing them that filling the gap has important consequences, emphasizing the relevance and significance of the research including its potential implications ([Wilson et al., 2024](#)) ([Gottlieb et al., 2018](#)) ([Lingard, 2015](#)).

A critical aspect of problem articulation is avoiding poorly defined problem statements, which represent one of the two most common weaknesses in introductions. These stem from the false perception that peer readers already know what the issue is and why the study to solve it is necessary ([Delvin et al., 2014](#)). Instead, authors must clearly note the broader theoretical, empirical and policy contributions and implications of their research ([Ezeifeke, 2021](#)).

The research problem should be presented in a way that identifies knowledge gaps while dealing with the objective and relevance of the study ([Schmitz, 2018](#)). This approach ensures that the gap identification goes beyond simply stating what hasn't been studied to demonstrating why addressing this particular deficiency matters for advancing scientific understanding or practical applications.

Evidence

(Sefcik et al., 2024)

[Writing Manuscript "Bookends": Strategies to Effectively Frame Science.](#)

"The introduction defines the problem to be addressed, identifies what is known and unknown about the problem, and states the study purpose. It begins broadly by introducing the area of interest, narrows to identify the specific focus and gap in knowledge, and finally ends with the aim of the present study, seamlessly leading to the methods and results sections."

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"establishing the scope, context and significance of the research, summarizing current understanding and background information about the topic"

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(Wilson et al., 2024)

[Journal recommended guidelines for survey-based research](#)

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(Gottlieb et al., 2018)

[Scholarship in Emergency Medicine: A Primer for Junior Academics Part I: Writing and Publishing](#)

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[Joining a conversation: the problem/gap/hook heuristic](#)

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(Delvin et al., 2014)

[How to Write a Scientific Paper: Practical Guidelines](#)

"The introduction has 3 main goals: to establish the need and importance of your research, to indicate how you have filled the knowledge gap in your field and to give your readers a hint of what they will learn when reading your paper. To fulfil these goals, a four-part introduction consisting of a background statement, a problem statement, an activity statement and a forecasting statement, is best suited. Poorly defined background information and problem setting are the 2 most common weaknesses encountered in introductions. They stem from the false perception that peer readers know what the issue is and why the study to solve it is necessary. Although not a strict rule, the introduction in clinical science journals should target only references needed to establish the rationale for the study and the research protocol."

(Schmitz, 2018)

[Susanne Kühn, Michael Kühn: Die Abschlussarbeit in den Life Sciences](#)

"Notes and guidelines for writing the introduction are summarised in chapter eight; inter alia, by dealing with the objective, relevance of the thesis, knowledge gaps, as well as with the research question and hypotheses, respectively."

Literature Review and Citation Management Strategies

Literature reviews in introductions should be selective rather than exhaustive, focusing only on studies directly relevant to contextualizing your research problem. Effective citation management involves using recent work (preferably within 10 years), ensuring accurate representation of sources, and maintaining logical connections between introduction and discussion sections. (6 sources)

The literature review component of scientific introductions requires a strategic approach that balances comprehensiveness with focus. **The broader context will be supported with the literature review**, which should include summaries of important research relevant to your study, particularly essential or groundbreaking studies about the research problem or key studies that refute or support your research ([Nor, 2020](#)). However, **a common mistake that authors make is to present a full review of the literature in their field**, when what is necessary is to mention only those aspects needed to contextualize the problem that your study is solving ([Nor, 2020](#)).

Effective citation practices emphasize recency and relevance. It is important to cite recent work, preferably within the last 10 years, from the available literature, and if available, citations should be from the country or region where the research is conducted ([Shankar, 2012](#)). The references cannot be exhaustive and can only be representative of the field ([Shankar, 2012](#)). This approach serves to introduce readers and reviewers to important literature about the subject area and the existing knowledge gap being addressed by the study ([Shankar, 2012](#)).

Citation management tools and databases can significantly streamline the literature review process. Using citation management software is one way to establish and maintain a working reference list, while full-text databases such as PubMed Central, NIH RePORT, and CDC Stacks can be useful when performing literature reviews ([Iskander et al., 2018](#)). Authors should familiarize themselves with databases accessible through their institution and seek assistance from reference librarians or interlibrary loan systems ([Iskander et al., 2018](#)).

Quality control and accuracy in citations are essential for maintaining scientific integrity. Authors should clearly understand the distinction between primary and secondary references and ensure they are knowledgeable about the content of any primary or secondary reference they cite ([Iskander et al., 2018](#)). Peer reviewers should evaluate whether the studies authors cite are rigorous and representative of the field, whether authors present a biased view of the literature, and whether there are excessive self-citations ([Proctor et al., 2025](#)).

Strategic citation planning involves considering the interconnection between introduction and discussion sections. Most references will be found in the introduction and discussion sections, as these are the most important parts of an article ([Podrigalo et al., 2023](#)) ([Jawaid et al., 2019](#)). If theories or studies mentioned in the introduction can explain the conclusions, then these sources should be used as a basis for discussing the results, and there is no need to repeat citations already used in the introduction ([Podrigalo et al., 2023](#)).

Evidence

(Nor, 2020)

[From Research and Practices into Journal Papers: Editor's Insight](#)

"Introduction is about placing your work into the broader research context, and then narrowing your focus to identify specifically what you plan to do in the paper -your research objectives. The broader context will be supported with the literature review. Therefore, introduction should include summaries of important and research which are relevant to your study. This is particularly important if there is an essential or groundbreaking study about the research problem or a key study that refutes or supports your research"

"However, a common mistake that authors make is to present a full review of the literature in their field. What is necessary is to mention those aspects that are needed to contextualize the problem that your study is solving."

(Shankar, 2012)

What do reviewers look for in an original research article?

"This section serves to introduce the reader and the reviewer to important literature about the subject area, and the existing knowledge gap which is being addressed by the study. The introduction serves to define and explain important terms and ideas relevant to the manuscript. The references cannot be exhaustive and can only be representative. It is important to cite recent work, preferably within the last 10 years, from the available literature. If available the citation should be from country or the region"

"The introduction should be well written and lead from one topic to another in an easy and well-defined sequence. The introduction is often the first section of the manuscript examined by the reviewer. A well written introduction produces a favorable impression about the manuscript."

(Iskander et al., 2018)

Successful Scientific Writing and Publishing: A Step-by-Step Approach

"Conducting a literature review is crucial to confirm the existence of the evidence gap that the planned analysis seeks to fill. Because literature searches are often part of applying for research funding or developing a study protocol, the citations used in the grant application or study proposal can also be used in subsequent manuscripts. Full-text databases such as PubMed Central (14), NIH RePORT (15), and CDC Stacks (16) can be useful when performing literature reviews. Authors should familiarize themselves with databases that are accessible through their institution and any assistance that may be available from reference librarians or interlibrary loan systems. Using citation management software is one way to establish and maintain a working reference list. Authors should clearly understand the distinction between primary and secondary references, and ensure that they are knowledgeable about the content of any primary or secondary reference that they cite."

(Proctor et al., 2025)

From novice to expert: preparing your peer review

"The peer reviewer should evaluate the strengths and weaknesses (or rigor) of the prior cited work, which the authors used to support their argument for undertaking the work presented in their manuscript. Peer reviewers should evaluate the introduction and Discussion as follows: • Is the present study the most logical next step, based on the presentation of the literature? • Are there any gaps in logic the authors must articulate to support their argument? • Are seminal or contradictory findings in the literature adequately presented and resolved? • Are the studies the authors cite rigorous (see definition of rigor) and representative of the field? • Are the authors presenting a biased view of the literature surrounding the scientific question? • Are citations correct? • Are there excessive self-citations?"

"These can include comments on the reporting of the methods (which may indirectly relate to rigor), as well as the tone, pacing, and content of the introduction and Discussion"

"It is important to ensure that references are accurately cited and, if not, noted in this section of your review. This includes verifying that the cited work directly supports the claims made and that there is no inappropriate self-promotion through citation. A thorough review of citations helps prevent bias and misrepresentation of sources in the manuscript."

(Podrigalo et al., 2023)

Duplicate references in the 'Introduction' and 'Discussion' sections of scientific articles on physical education and sports

"Research on various aspects of preparing the "Discussion" section presents specific recommendations on structure, language, and sequence of material presentation (Kearney, 2017)(Conn, 2017)(Höfler et al., 2018)(Al-Shujairi et al., 2020)(Corlett, 2007)[6]. Many studies support the idea of a logical interconnection between sections of a scientific article. For instance, Jawaid et al (Jawaid et al., 2019) assert that the "Introduction" and "Discussion" sections are the most important parts of an article. Therefore, most references will be found in these sections"

"In the study by Vieira et al. (Vieira et al., 2019), it is recommended to start the discussion by recalling important information included in the "Introduction" and/ or "Material and Methods" sections. This approach allows for contextualizing the research"

"Goulston provides a detailed explanation of the interrelation between the content of the "Introduction" and "Discussion" sections [10]. The author notes that if theories or studies mentioned in the introduction can explain the conclusions, then these sources should be used as a basis for discussing the results. This is corroborated by the research of Bavdekar [11], who points out that the "Discussion" section should be written purposefully and directly address the research question raised in the "Introduction" section. It is also recommended to compare the research results with those of recent, relevant highquality studies. Wordvice [12] advises that in the "Discussion" section, authors should demonstrate how their research findings fit into the broader scientific literature reviewed in the "Introduction" section"

"Enago Academy [13] notes that studies mentioned in the literature review should only be used as support and evidence for the research"

"For instance, Nundy [15] recommends using the same key concept in the discussion that was introduced in the introduction. The authors believe that there is no need to repeat citations already used in the introduction."

(Jawaid et al., 2019)

How to write introduction and discussion

"Scientific writing has a proper format. The anatomical structure or important components of an original article consists of a structured abstract (usually in four sub-headings i.e. objective, methods, results and conclusions with appropriate key words), introduction, methodology, results and discussion. It is summarized by the acronym IMRAD wherein: I stands for Introduction, M for Methods, R for Results, A for And, D for Discussion. This manuscript describes the information which has to be included in the Introduction and Discussion sections of an original article."

Common Mistakes to Avoid in Introduction Writing

Common introduction mistakes include starting with excessive citations, writing overly long introductions that lose focus, using outdated references for current situations, and making weak claims about research being "understudied" or "important" without letting evidence speak for itself. (3 sources)

- **Starting with excessive citations in opening sentences** - Authors frequently write whole series of citations in the first sentences as if they demonstrated their knowledge of the subject matter, but this should be avoided. Instead, start with your own comment, then support your view with citations ([Lamanauskas, 2019](#)).
- **Writing overly long introductions that lose focus** - Mistakes are more related to very long introductions when the logical sequence of the text and the real problem to be highlighted by the research are lost. This makes the text tiresome to read and loses its introductory characteristics ([Silveira et al., 2022](#)).
- **Raising issues not covered in the article** - Another common mistake is to raise issues that will not be covered in the article, which confuses readers about the actual scope and objectives of the study ([Silveira et al., 2022](#)).
- **Excessively detailing previous studies** - Providing too much detail about what was observed in previous studies makes the text tiresome to read and loses its introductory characteristics ([Silveira et al., 2022](#)).
- **Using outdated references for current situations** - Citing current situations using old references is a recurrent mistake in scientific studies that undermines the relevance and timeliness of the research context ([Silveira et al., 2022](#)).
- **Making weak claims about research importance** - Beware of asserting that research questions are "interesting," "important," or "understudied." Let the reader conclude that they are interesting or important. As for "understudied," it makes a weak case as topics may be understudied because they are unimportant ([Carandini, 2022](#)).
- **Failing to end with clear research aims** - The introduction should end with a clear formulation of the research aim and questions, presenting a brief overview of what is supposed to be solved during the reported original research ([Lamanauskas, 2019](#)).
- **Exceeding optimal length** - Strive for brevity, ideally keeping the introduction below 500 words to maintain reader engagement and focus ([Carandini, 2022](#)).

Evidence

(Lamanauskas, 2019)

SCIENTIFIC ARTICLE PREPARATION: A COMPREHENSIVE INTRODUCTION

"The beginning of the introduction is another very common misconception (Table 3). Authors frequently write the whole series of citations and/or references yet in the first sentences as if they demonstrated their knowledge of the subject matter. This should be avoided. On the contrary, it makes much more sense and purpose when a researcher starts with his/her own comment, then supports his/her view with citations and/or other adaptations. In other words, an appropriate beginning of the introduction is clearly visible and can be referred to as an 'opening paragraph'"

"it should be acknowledged that scientific texts are produced for experts in a particular field of science, and therefore there is no

need for terms/concepts to be explained and clarified each time: their meaning is clear and conveyed. On the other hand, researchers suggest it is sometimes necessary to explain particular terms or words in the papers, which is the case to do so (Kvašytė & Macienė, 2010)"

"As a rule, the introduction ends with a clear formulation of the research aim and question (s). Thus, the introduction usually presents researchers' brief overview of what is supposed to be solved during the reported original research."

(Silveira et al., 2022)

Guide for scientific writing: how to avoid common mistakes in a scientific article

"The introduction should be the last section to be written. Thus, authors will have greater objectivity in the relevant subjects to introduce the research topic. It should introduce the subject and provide reasons to justify the objective and relevance of the study. The introduction also presents the research problem and demonstrates knowledge gaps. A logical sequence should be established from the most general to the most specific aspects of the study (inverted triangle)"

"Mistakes in this section are more related to a very long introduction when the logical sequence of the text and the real problem to be highlighted by the research are lost. Another common mistake is to raise issues that will not be covered in the article. Excessively detailing what was observed in previous studies makes the text tiresome to read and loses its introductory characteristics. Citing current situations using old references is also a recurrent mistake in scientific studies"

"The introduction must contain: 1) Scenario (general information establishing the field of study) 2) Background (presentation of relevant previous studies) 3) Knowledge gap 4) Research question (concise definition of objective/hypothesis)"

(Carandini, 2022)

Some Tips for Writing Science

"Let's start here in the first paragraph, which should set up the broad area of inquiry and ideally open with a vivid sentence"

"The rest of the Introduction is there to pose specific questions in the context of the literature and as objects of debate. These questions are the pillars of the paper: pose them in the Introduction, answer them in Results, and discuss your answers in Discussion"

"When presenting the questions, beware of asserting that they are "interesting," "important," or "understudied." Let the reader conclude that they are interesting or important. As for "understudied," it makes a weak case: they may be understudied because unimportant"

"The last paragraph of the Introduction usually describes the main results. Mention the methods, especially if they are novel or provide new avenues into a problem, and summarize the take-home messages"

"Strive for brevity, ideally keeping the Introduction below 500 words."